



SARDAR PATEL UNIVERSITY

T.Y.B.C.A. SEMESTER – V (CBCS-NC) EXAMINATIONS – 202

US05FBCA01: Operation Research

Tuesday, 22nd March, 2022

Time: 02.00 PM to 04.00 PM

Max. Marks: 70

Q-1. Multiple choice questions

[10]

1. A physical model is an example of _____.
a. An iconic model b. An analogue model
c. A verbal model d. A mathematical model
2. If for a given solution, a slack variable is equal to zero, then _____.
a. It lies on the critical path. b. It is a dummy activity
c. The project progressing well d. None of these.
3. A constraint in an LP model restricts _____.
a. value of the objective function b. value of the decision variable
c. use of the available resources d. all of the above
4. In the optimal simplex table $c_j - z_j = 0$ value indicates _____.
a. unbounded solution b. cycling
c. alternative solution d. None of these
5. The solution to a transportation problem with m-rows and n-columns is feasible if number of positive allocations are _____.
a. $m+n$ b. $m \times n$ c. $m + n - 1$ d. all of the above.
6. The method used for solving an assignment problem is called _____.
a. reduced matrix method. b. MODI method
c. Hungarian method d. none of the above
7. North West Corner Refers _____.
a. top -right corner b. top left corner
c. bottom left corner d. none of them
8. In non degenerated solution number of allocated cell is _____.
a. equal to $m+n+1$ b. not equal to $m+n+1$
c. equal to $m+n-1$ d. not equal to $m+n-1$
9. The slack for an activity is equal to _____.
a. LF-LS. b. EF-ES c. LS-ES. d. None of the above.
10. The objective function for a L.P model is $3x_1 + 2x_2$, if $x_1 = 20$ and $x_2 = 30$, what is the value of the objective function?
a. 0 b. 50 c. 60 d. 120

Q-2. Fill in the Blanks

[08]

1. Linear programming problem involving only two variables can be solved by _____.
2. If the value of the objective function z can be increased or decreased indefinitely, such solution is called _____.
3. Activity which starts only after finishing other activity is called _____.
4. The linear function of the variables which is to be maximized or minimize is called _____.
5. Processing time M_{ij} 's are _____ of order of processing the jobs.
6. The penalty in VAM represents difference between _____ costs of respective row/column.
7. Activity which does not require any resources or time is called _____.
8. _____ operation is carried out on a machine at a time.

Q-3. Short Questions (Attempt any ten).

[20]

1. Write the definition of solution, basic solution.
2. How will you construct a mathematical model?
3. Write the definition of operation research.
4. Write the advantages of LPP.
5. How will you plot inequalities of a LPP?
6. What is artificial variable?
7. List out phases of operation research.
8. Define terms: Events, Merge event.
9. What is Assignment problem?
10. What is dynamic programming problem?
11. Write steps for Matrix Minima method.
12. Explain: 1) analytical model 2) simulation model.

Q-4. Long Answer Questions: (Any 4 out of 8)

[32]

1. Write the steps for solving Linear Programming Problem by Graphical method. State its limitations
2. What is model? List out the model.
3. Maximize $Z = 30x_1 + 20x_2$
s.t. $3x_1 + 3x_2 \geq 40$
 $3x_1 + x_2 \geq 40$
 $2x_1 + 5x_2 \geq 44$
 $x_1, x_2 \geq 0$

4. Write the steps of the algorithm for solving LPP by the Simplex method.
5. Determine an IBFS by Least Square method.

Source	D1	D2	D3	D4	Supply
S1	19	30	50	10	7
S2	70	30	40	60	9
S3	40	8	70	20	18
Demand	5	8	7	14	

6. Write the step to solve assignment problem by Hungarian method.
7. A project schedule has the following characteristics. Construct the PERT network and find the critical path and time duration of the project.

Activity	1 - 2	1 - 4	1 - 7	2 - 3	3 - 6	4 - 5	4 - 8	5 - 6	6 - 9	7 - 8	8 - 9
Time	2	2	1	4	1	5	8	4	3	3	5

8. State the rules for drawing network diagram.



